Orbius: Multi-Agent BRD Studio

HackSprint

24-Hour Hackathon

17-18 OCTOBER

VENUE DETAILS
Manipal Academy of Higher Education (MAHE)

SATURDAY
09 A.M. Onwards

YOUR NAME: Adhyayan Dimri

TEAM NAME: LocalHost

TRACK NO: T.04 PS-42

REGISTER NOW

Team Size
1-4 Members

ORGANIZED BY:



Solution

The problem Requirements live in emails, calls, PDFs, sheets and wireframes, and they contradict each other.

Orbius Reads it all in parallel, flags contradictions, and keeps one validated BRD alive.

1 Ingest

PDF, DOCX, CSV/XLSX, email, images, voice notes

2 Extract in parallel

Text, Vision, Document, Budget, Risk and Ops agents

3 Validate

Dedupe, drop ungrounded claims, flag conflicts

4 Living BRD + actions

10-section BRD, diagrams, draft emails, alerts

Orbius in action: the Q3 vs October launch conflict

SOURCE A | Executive Memo (VP of Product)

"Our launch window is locked to Q3."

SOURCE B | Field Operations & Finance

"We cannot staff stores before October."

ORBIUS

Flags it, asks which window is canonical, and a human picks Keep Q3 or Keep Q4.

Prototype Link

Report

GitHub Repo

Tutorial Video

Tech Stack & Architecture

Frontend

- React 18 + Vite
- Mermaid.js diagrams
- Web Speech API voice notes
- Live updates over server-sent events

Backend

- FastAPI + Uvicorn (Python)
- Pydantic v2 typed schemas
- Sections streamed via SSE
- Agents run in parallel threads
- Docker and Kubernetes

AI layer

- Google Gemini (google-genai)
- 3-tier model router
- Offline fallback engine
- Gemini audio transcription

Data and ingestion

- SQLite: users, tasks, alerts, sessions
- pypdf, openpyxl, Pillow, DOCX parser
- Chunks keep source ID and page

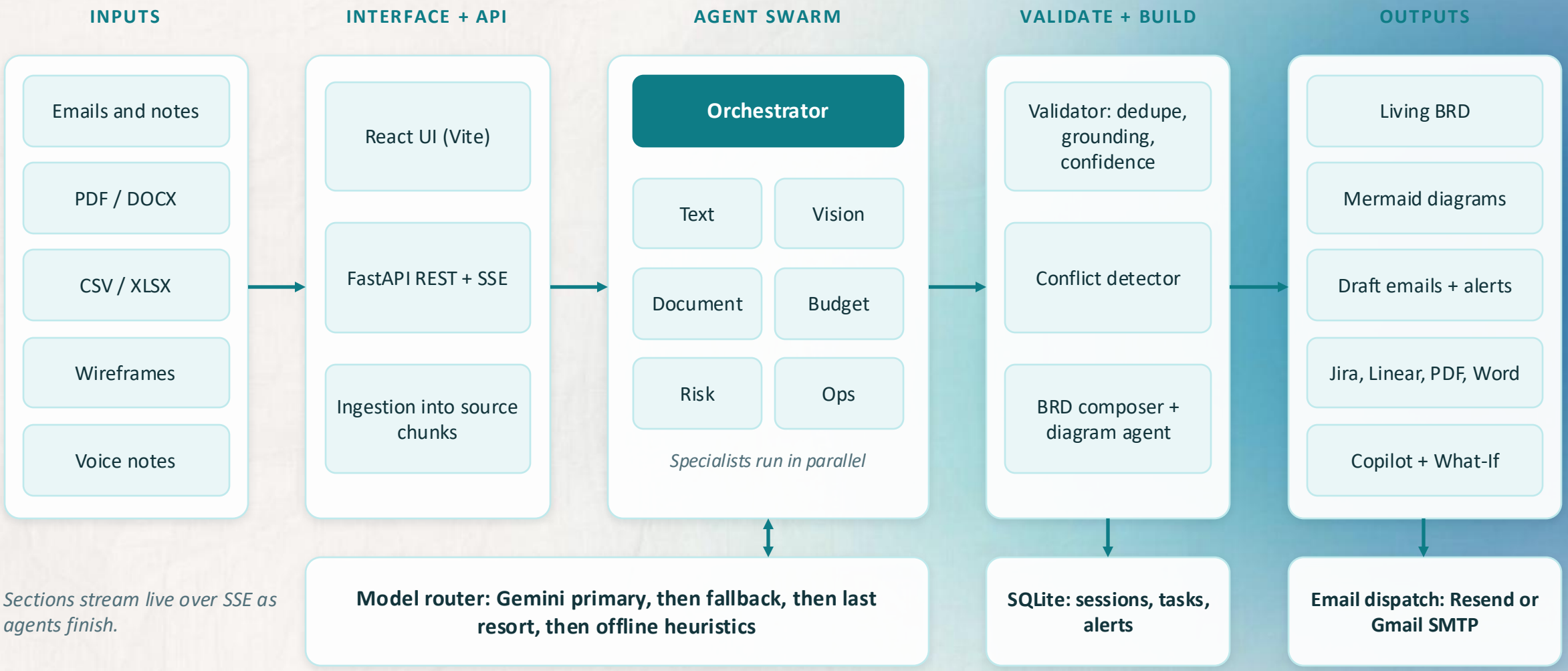
Exports

- PDF (ReportLab) and Word (.docx)
- Jira CSV with Gherkin criteria
- Linear JSON backlog
- Markdown

Ops and delivery

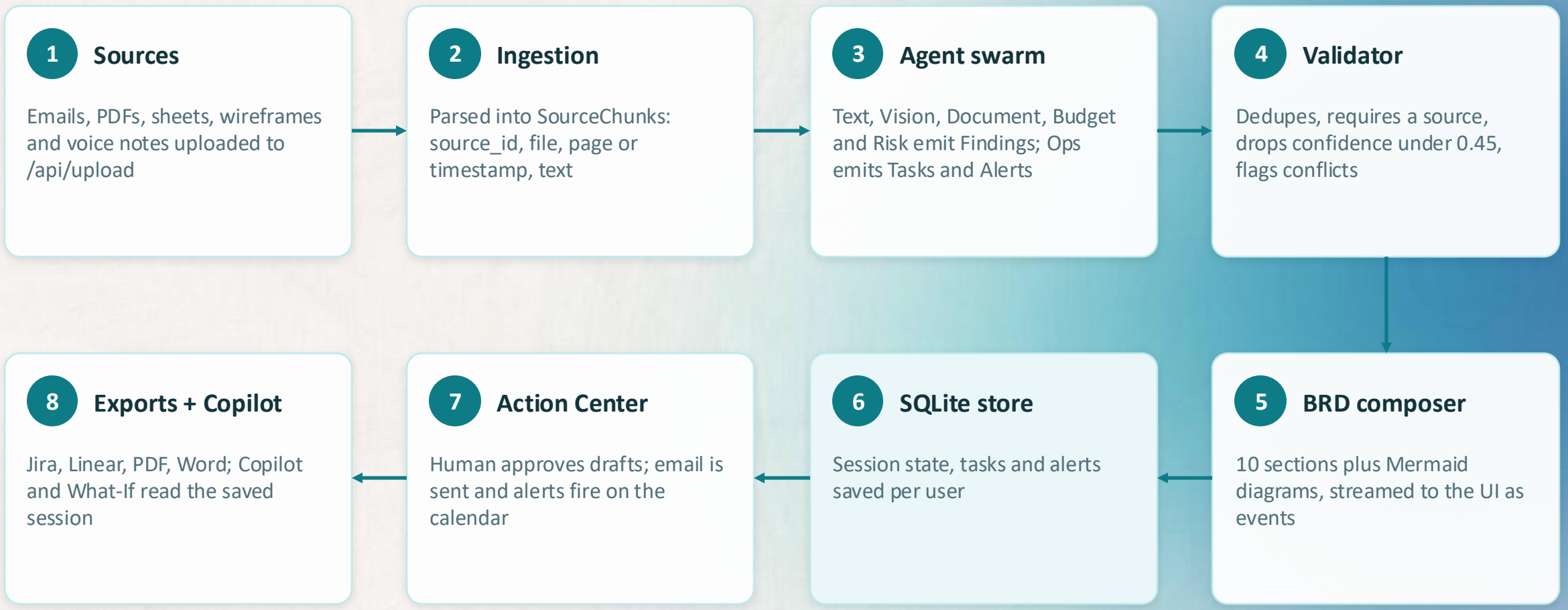
- Resend API or Gmail SMTP dispatch
- Simulated clock for alert firing
- Human approves before anything sends

System Architecture



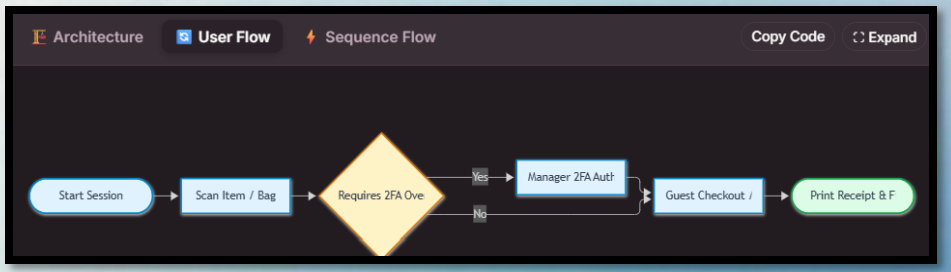
Sections stream live over SSE as agents finish.

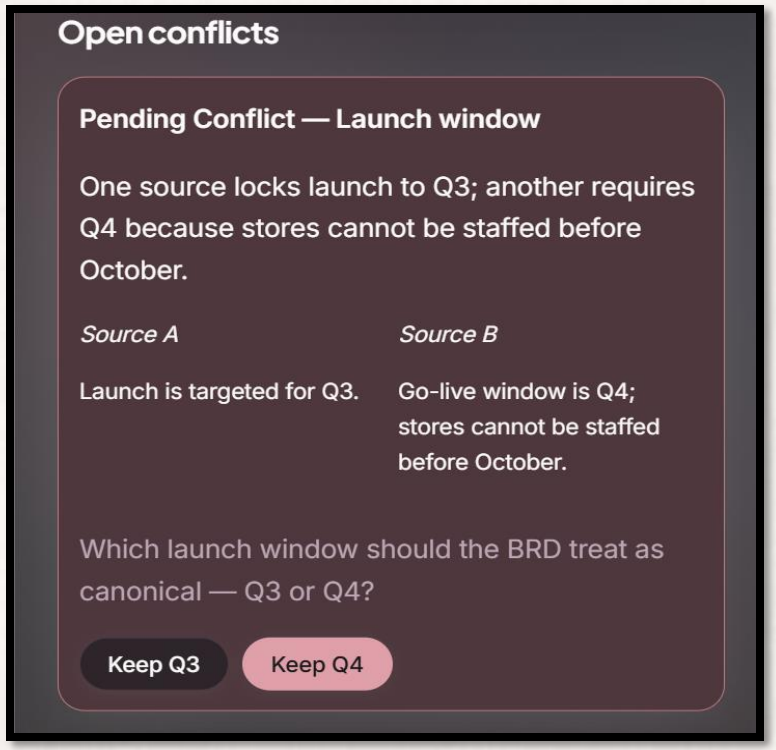
Data Flow Diagram

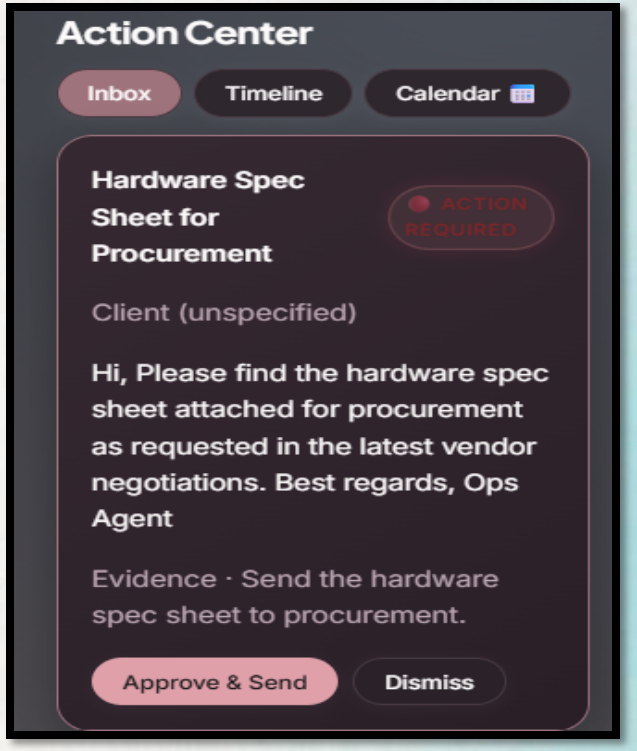


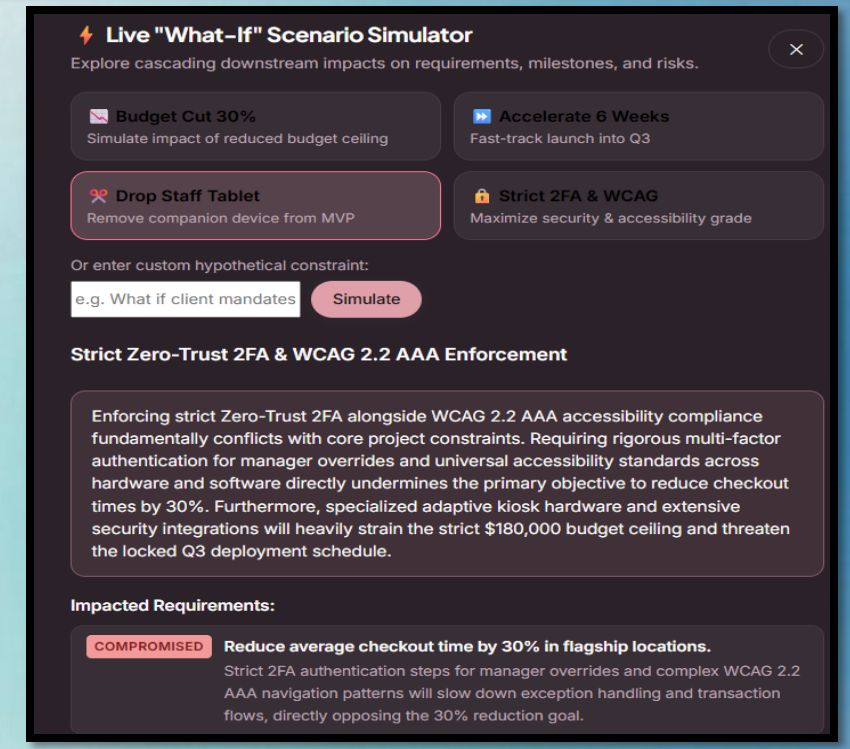
Every Finding keeps its source_id, exact source span, agent and confidence score, so evidence stays traceable.

Screenshots of Our Project









Living BRD

Sections with evidence drawer and the Q3 vs Q4 conflict card

Action Center

Draft emails with Approve & Send, and the urgency calendar

Diagrams and What-If

Architecture diagram, kiosk prototype or scenario simulator

Others

Why it stands out

- Every requirement traced to its source text and confidence
- Conflicts surfaced for a human decision, not silently merged
- Requirements and follow-up tasks kept separate by the Ops Agent

Beyond the document

- Mermaid architecture, flow and sequence diagrams
- Clickable kiosk prototype and Gherkin test scenarios
- Jira CSV, Linear JSON, PDF and Word exports

Think with your BRD

- Grounded Copilot with citation chips
- What-If: budget cut, faster launch, dropped scope
- Impact ripple graph and Live Meeting interrupter

Resilience and next steps

- Gemini failover chain plus offline engine
- Next: Gmail OAuth, real authentication, vector retrieval
- Next: push straight to Jira and Linear APIs

Demo data: 5 stakeholder personas, one $180k budget cap, one Q3 vs October conflict. Team: [names]